

NITISH KUMAR SAH

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Work Experience

Saptang Labs – Computer & Network Security

Dec 2025 – Present

Software Engineer Intern

Remote

- Developed **50+ secure REST API endpoints** using FastAPI with **RBAC**, **user management**, **input validation**, and **activity logging**, ensuring auditability of **100% critical user actions**.
- Designed a **multi-factor authentication system** (Email OTP + TOTP) and built an **asynchronous email pipeline** (API → Redis Queue → Worker → SMTP) with **retry handling** and **non-blocking** request flow.
- Optimized **database queries** and implemented **pagination** for high-volume APIs, reducing average response time by **30–40%** and improving large dataset handling.

Projects

CampusSentinel – AI Campus Surveillance (Runners-Up at IIT Guwahati)

[GitHub](#) - [Demo](#)

Django REST Framework, FastAPI, Pydantic, JWT, PostgreSQL, Redis

- Designed a **relational PostgreSQL schema** with **12+ indexed tables** to correlate multi-source campus events.
- Built a **microservice backend** (Django REST Framework+FastAPI) with **10+ endpoints** and JWT authentication.
- Implemented **ML model integration pipelines** within FastAPI services to process and serve structured responses.

MediMind – AI-Powered Medical Assistant Platform

[GitHub](#) - [Live](#)

Django REST Framework, FastAPI, Pydantic, PostgreSQL, LangChain

- Developed **15+ REST APIs** with **JWT authentication** and **Pydantic-based input and output validation**.
- Designed a **normalized PostgreSQL schema** with **relational models** for users, patients, and prescriptions.
- Integrated **LLM-based AI services** into FastAPI with **schema-validated outputs**, enabling reliable and structured medical response generation.

Technical Skills

- Languages:** Python, C++, JavaScript, SQL
- Backend & API Security:** REST API Design, Authentication (JWT, MFA), RBAC, Input Validation
- Databases:** PostgreSQL, MongoDB, MySQL, Redis (Caching, Queues)
- Frameworks:** FastAPI, Django REST Framework, Pydantic, Motor
- Systems & DevOps:** Async Processing, Background Workers, Docker, Git, GitHub

Education

National Institute of Technology, Rourkela

Aug 2023 – Present

B.Tech in Biotechnology

Rourkela, Odisha

DAV Public School, Hehal, Ranchi

2022

CBSE, Science (PCM)

Ranchi, Jharkhand

Achievements/Certifications

- Runners-Up, Ethos 2025, IIT Guwahati** [Certificate](#)
- 2nd Runners-Up, Hackfest'25, IIT-ISM Dhanbad** [Certificate](#)
- Competitive Programming:** Codeforces Pupil (Rating: 1200+) | CodeChef (Rating: 1500+)
- Supervised Machine Learning: Regression and Classification** – DeepLearning.AI [Certificate](#)
- Advanced Learning Algorithms** – DeepLearning.AI [Certificate](#)

Extracurricular Activities

Pantomime - Dramatics Club

Mar 2024 – Present

Treasurer

NIT Rourkela

Webwiz - Web Development Club

Jan 2025 – Present

Technical Team Member

NIT Rourkela

Innovision - Technical Fest

Sep 2024 – Nov 2024

Coordinator

NIT Rourkela